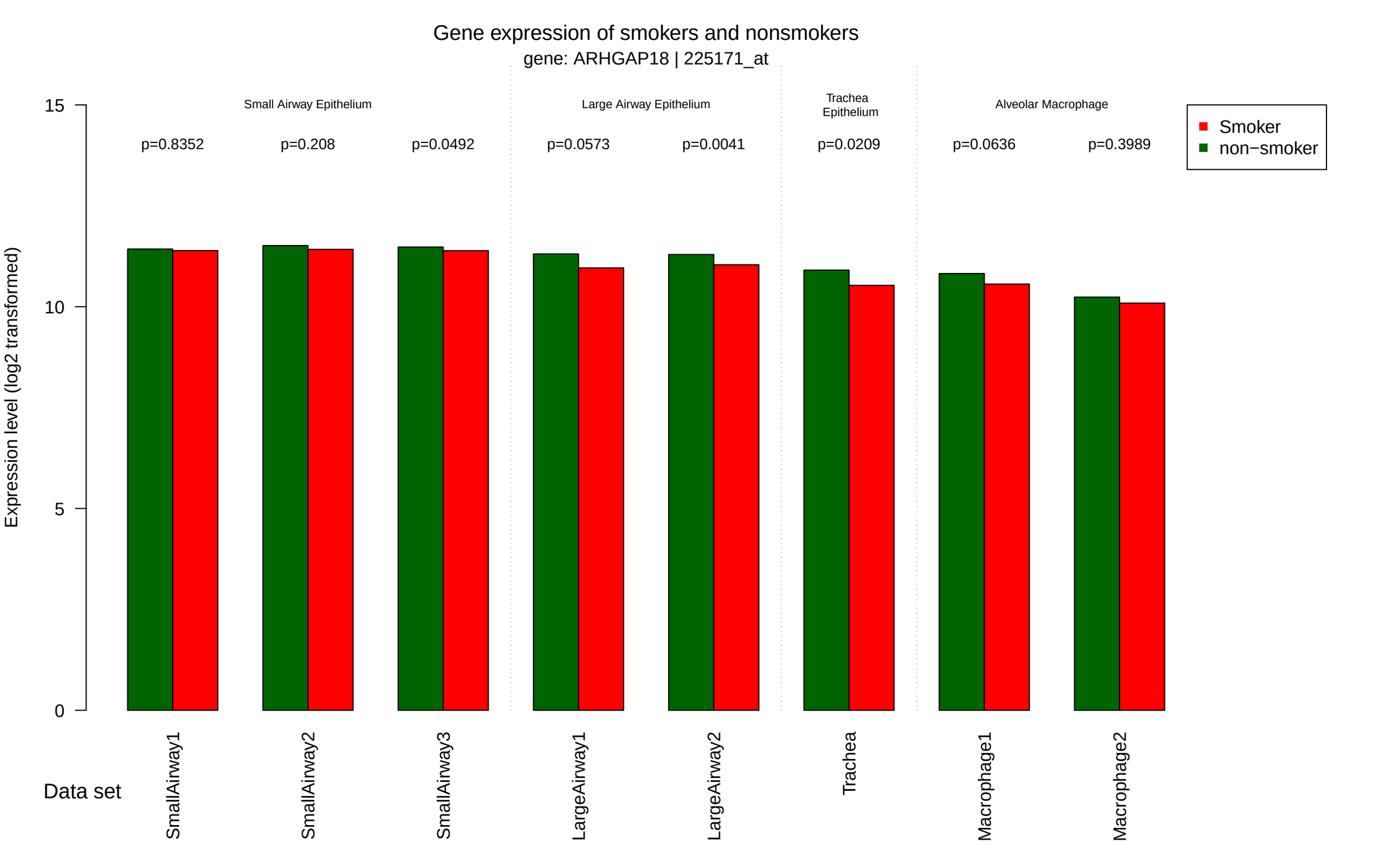
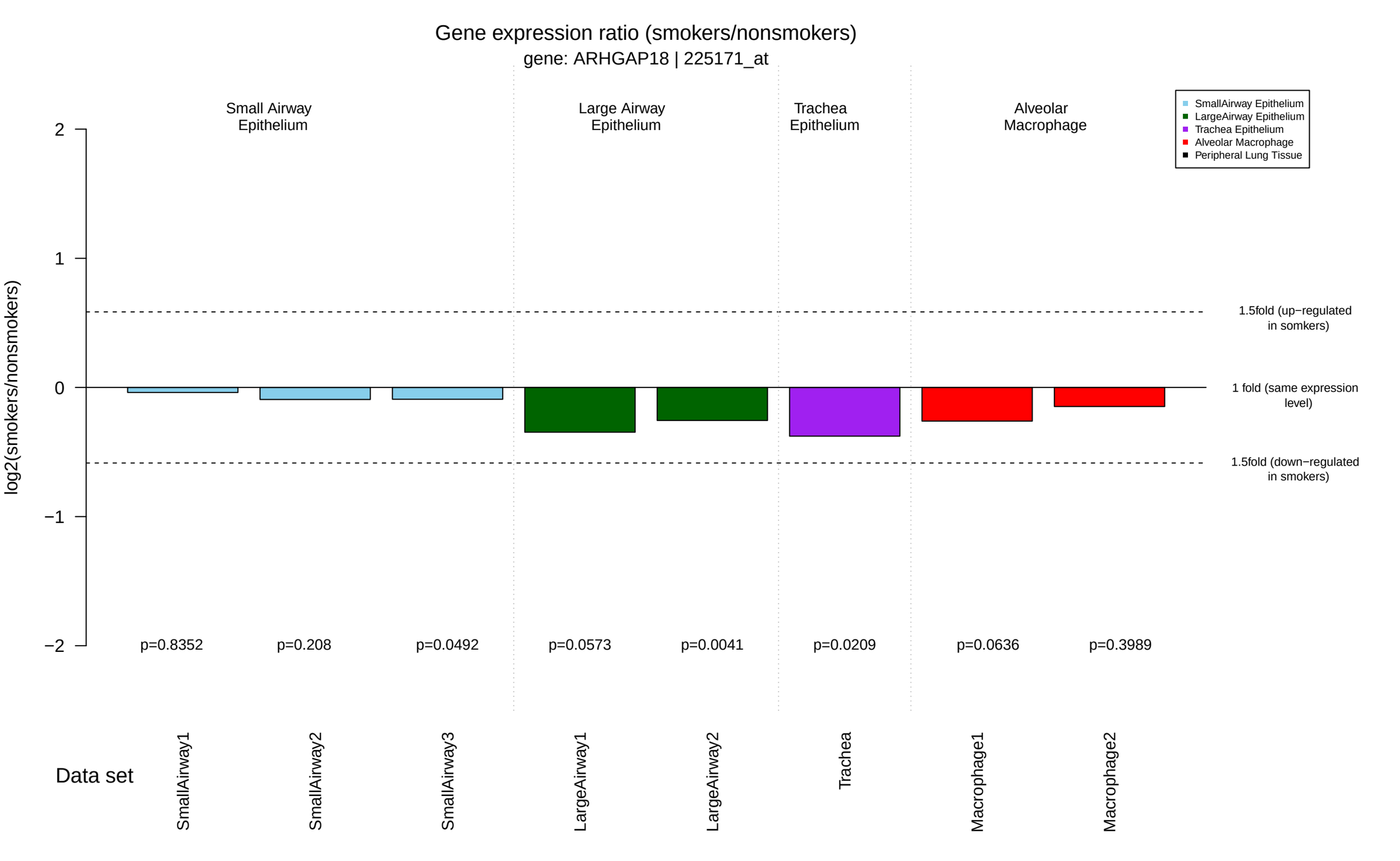
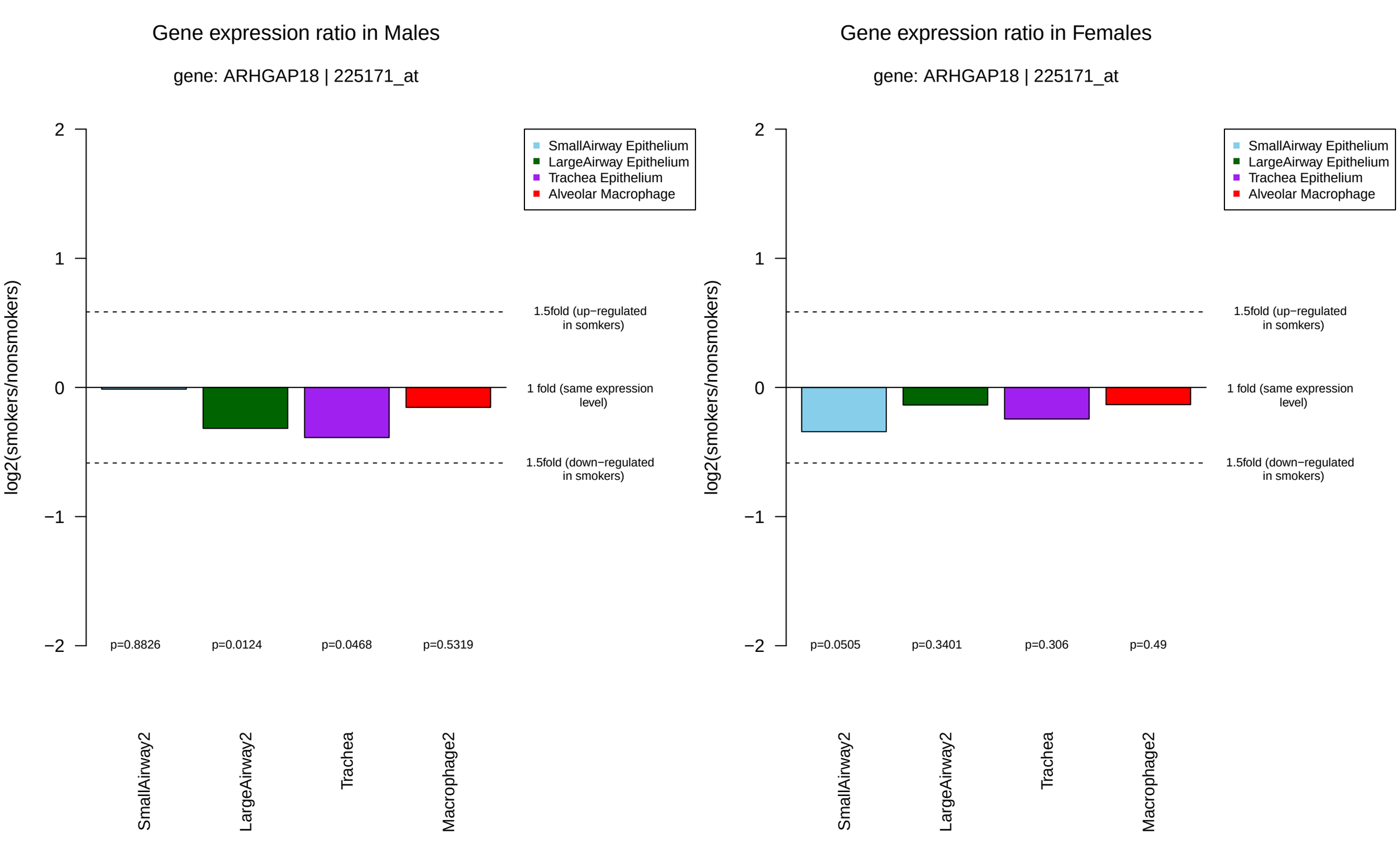
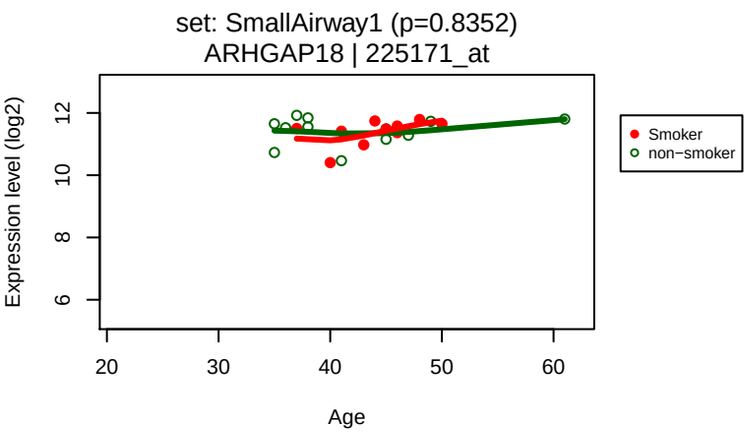
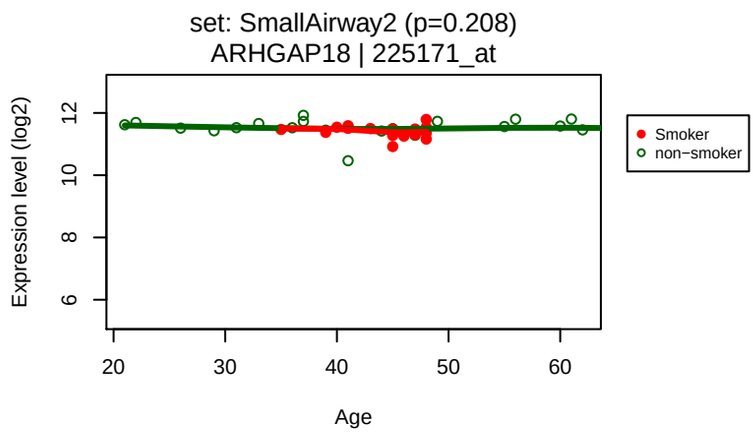
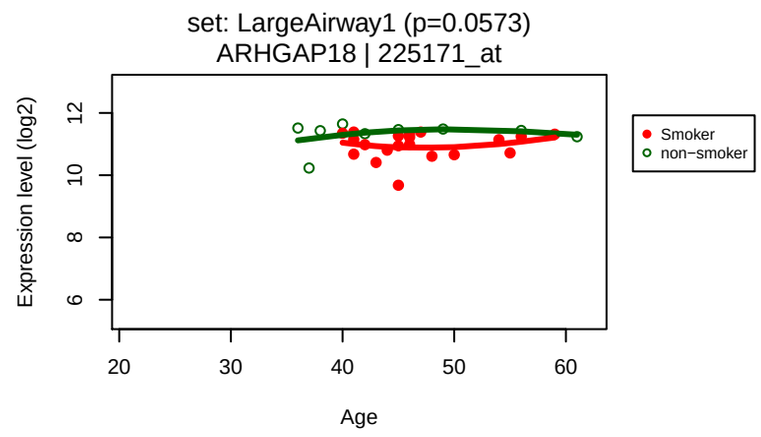

Gene expression of smokers and nonsmokers
gene: ARHGAP18 | 225171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8352
p=0.208
p=0.0492
p=0.0573
p=0.0041
p=0.0209
p=0.0636
p=0.3989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

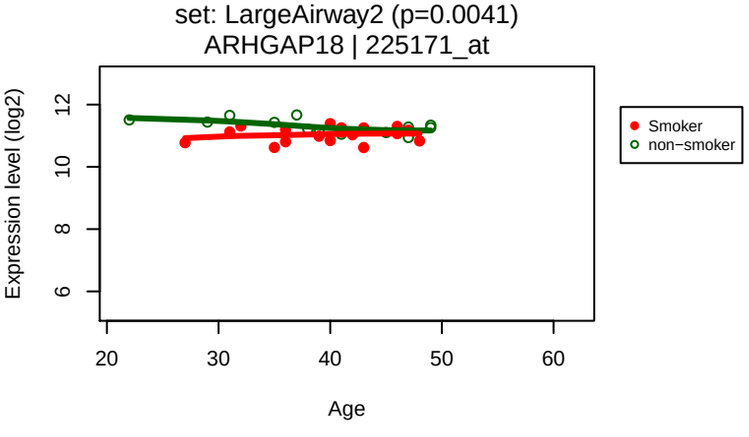

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP18 | 225171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8352
p=0.208
p=0.0492
p=0.0573
p=0.0041
p=0.0209
p=0.0636
p=0.3989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

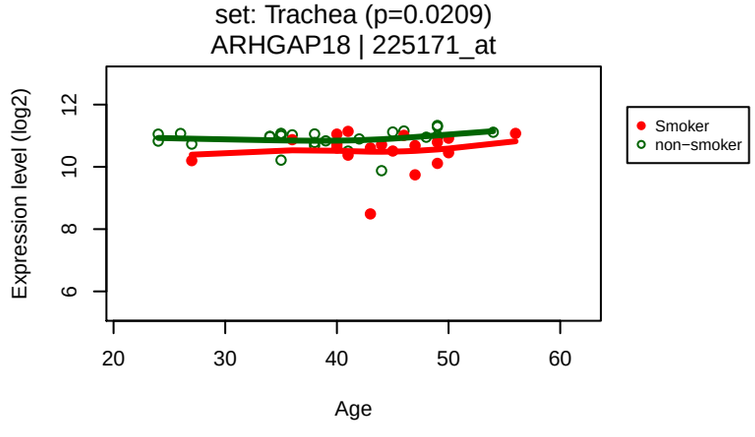

Gene expression ratio in Males
gene: ARHGAP18 | 225171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8826
p=0.0124
p=0.0468
p=0.5319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP18 | 225171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0505
p=0.3401
p=0.306
p=0.49
SmallAirway2
LargeAirway2
Trachea
Macrophage2

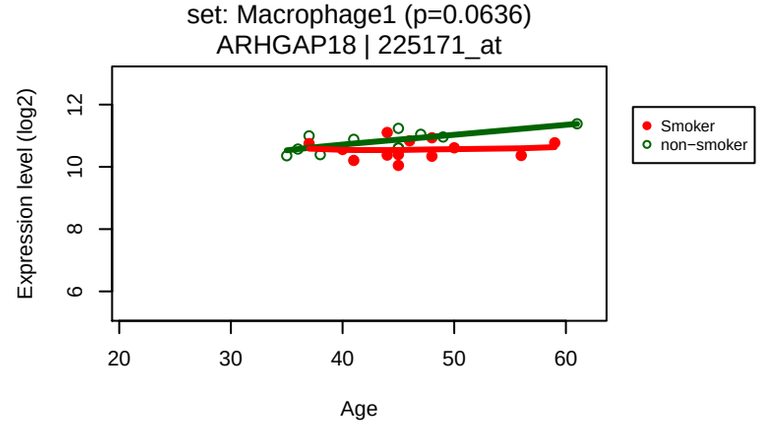

set: SmallAirway1 (p=0.8352)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker

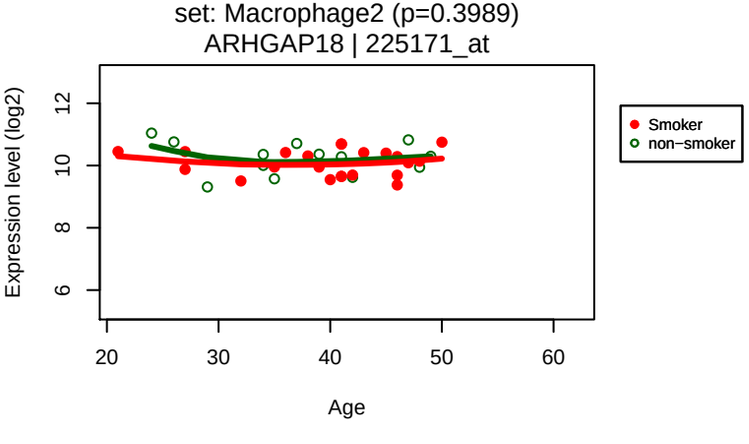

set: SmallAirway2 (p=0.208)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker

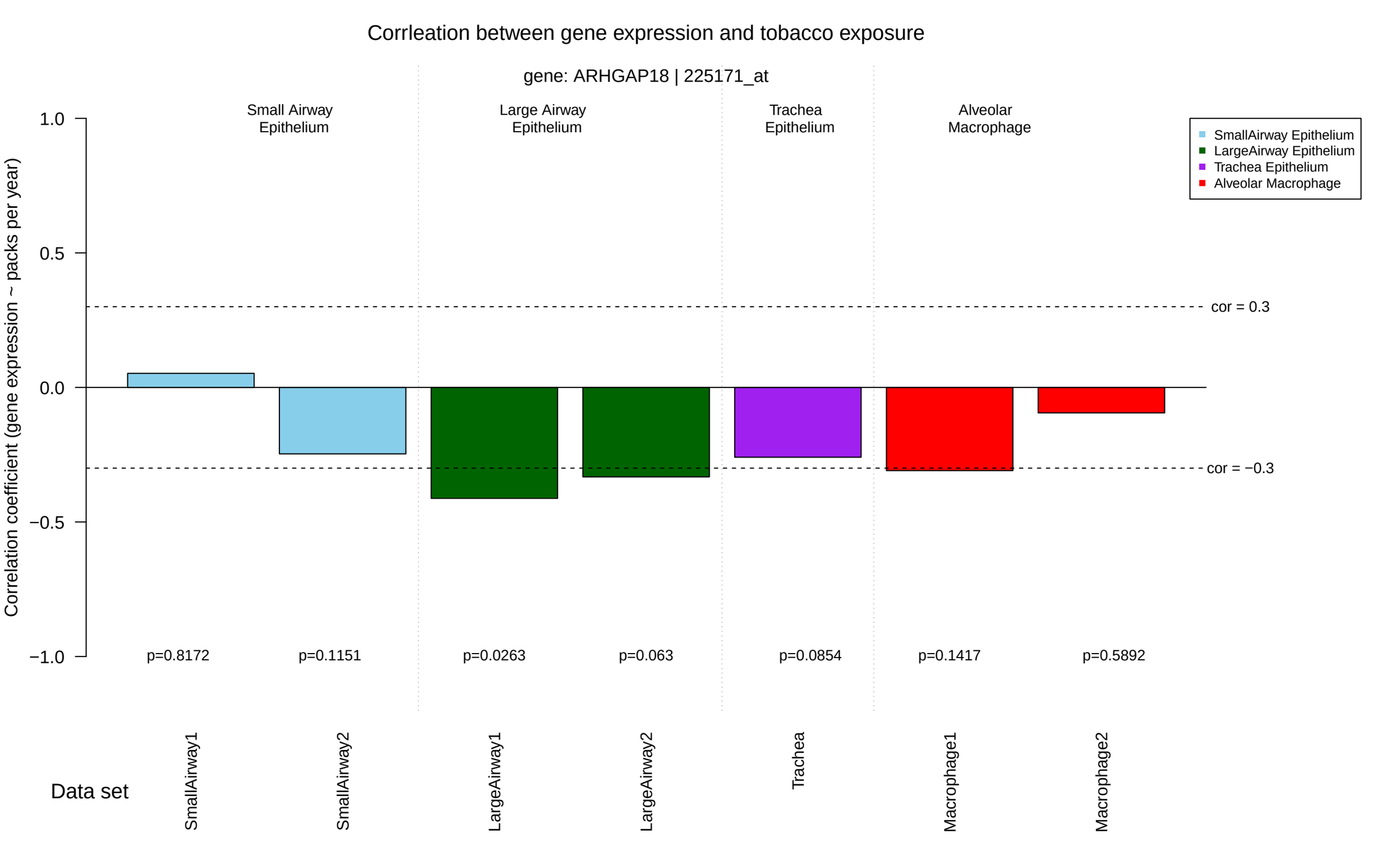

set: LargeAirway1 (p=0.0573)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0041)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0209)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0636)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3989)
ARHGAP18 | 225171_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP18 | 225171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8172
p=0.1151
p=0.0263
p=0.063
p=0.0854
p=0.1417
p=0.5892
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2